

Hans-Peter Marshall

Cryosphere Geophysics And Remote Sensing (CryoGARS) group
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Education

- 2005 Ph.D, Civil Engineering (Geotechnical), GPA: 3.9/4.0
Institute of Arctic and Alpine Research, University of Colorado at Boulder
Thesis: *Snowpack spatial variability: towards understanding
its effect on remote sensing measurements and snow slope stability*
- 1999 B.S., Physics with honors, Geophysics Minor (1st in UW history), GPA: 3.6/4.0
University of Washington
Thesis: *Wet snow densification*
- 1994 Nathan Hale High School, Seattle, Washington, GPA: 3.9/4.0

Professional Experience

- 08/2013-present Associate Professor, Dept. of Geosciences, Director of CryoGARS, Boise State University
- 06/2018-present Project Scientist, NASA SnowEx Mission (20+ organizations, 100+ participants)
- 07/2007-03/2020 Geophysical Engineer (Expert Consultant), U.S. Army Corps of Engineers
Cold Regions Research and Engineering Lab (CRREL), Hanover, New Hampshire
- 08/2008-07/2013 Assistant Professor, Professor, Department of Geosciences and Center for Geophysical
Investigation of the Shallow Subsurface (CGISS), Boise State University
- 04/2006-08/2008 Research Associate, Inst. of Arctic and Alpine Research (INSTAAR)
University of Colorado at Boulder
- 12/2004-04/2006 Research Geophysical Engineer (GS-11), Student Career Experiences Program (SCEP)
U.S. Army Corps of Engineers CRREL, Hanover, New Hampshire & Fairbanks, Alaska
- 01/2004-07/2004 Invited visiting scientist, Swiss Federal Institute for Snow and
Avalanche Research (SLF), Davos, Switzerland
- 07/2000-12/2005 Research Assistant, INSTAAR, University of Colorado at Boulder
- 11/2002-12/2003 Avalanche Instructor, Alpine World Ascents, Boulder, Colorado
- 06/1995-07/2000 Undergraduate Research Assistant, Glaciology Group, University of Washington
- 08/1999-03/2000 Volunteer high school teacher, Old Crow, Yukon Terr., Canada
- Summers 1993-94 Roofer, Seattle, WA

Research Interests

Snow science and glaciology, using geophysics, engineering, and remote sensing
Spatial variability of snow, snow and ice mechanics, snow slope stability, meltwater dynamics
Computational statistics, geostatistics, compressed sensing, instrumentation development

Honors and Awards

2020	BSU Faculty Excellence Award
2019	2018 WRR Editors' Choice Award: Hedrick et al., 2018
2018	NASA Robert H. Goddard Team Award: SnowEx Year 1 Organizing Team
2017	Incoming Chair of the International Snow Working Group for Remote sensing (ISWGR)
2012	Timeless Award, University of Washington, 1 of 150 College of Arts and Sciences alumni awarded from UW history on 150th yr anniv. (1861-2011), 1 of 3 from Physics Department
2010-13	New Investigator Program Award, NASA, 1st in Idaho history
2010	Young Scientist Award, Cryosphere Section, American Geophysical Union
2009	Top Reviewer Award, Cold Regions Science and Technology (CRST)
2008	U.S. Young Scientist Award, International Union of Radio Science (URSI)
2002-05	N.A.S.A. Earth Systems Science Graduate Fellowship
2003	Backcountry Access Avalanche Education Support Grant
2003	Arctic Research Consortium of the U.S. (ARCUS) Award for Arctic Research Excellence, Hon. Mention
2002	University of Colorado Graduate School Student Travel Grant
2001-02	N.S.F. K-12 Engineering Outreach Fellowship
2002	Beverly Sears Graduate Research Grant (University of Colorado)
2001	American Avalanche Association Graduate Research Grant
2000	Li Tze Fong Memorial Fellowship, University of British Columbia (declined)
1999	University of Washington Bonderman Honors Travel Fellowship
1999	Volunteer of the Year, Arctic Canada, Frontiers Foundation
1997-99	University of Washington Mary Gates Undergraduate Research Grant
1999	Featured speaker, University of Washington Undergraduate Research Symposium
1995-98	N.A.S.A. Space Grant Undergraduate Research Program
1999,2001	N.S.F. Graduate Fellowship Program - Honorable Mention
1994-95,97-99	Dean's List, University of Washington
1994-95	U.W. Undergraduate Scholar Award (1994-95)
1994	Washington Athletic Club Student Athlete of the Year, Nathan Hale High School

Student Awards

2018-2020	Joel Gongora, Ph.D. student <i>NASA Earth Systems Science Fellowship (~50 awarded nationwide)</i>
2018	Tate Meehan, Ph.D. student <i>Idaho Space Grant Consortium, NASA EPSCoR graduate fellowship</i>
2012-2014	Scott Havens, Ph.D. student <i>NASA Earth Systems Science Fellowship (~50 awarded nationwide)</i>
2012	Andrew Hedrick, M.S. student <i>American Avalanche Association Graduate Student Grant</i>
2012	Scott Havens, Ph.D. student <i>1st place poster NSF/NASA EPSCoR annual meeting</i>
2012	Kris Burch, Lloyd Lowe II, Kolton Drake, Undergrads, <i>Best Senior Project, BSU College of Engineering</i>
2003	Adam Maerz, High School student <i>Colorado State Science Fair Finalist</i>

Selected Refereed Publications [peer-reviewed publications=63, h-index=27, i10-index=50, citations=2192]

- 2021 **H.-P. Marshall**, Deeb, E., Forster, R., Vuyovich, C., Elder, K., Hiemstra, C., and Lund, J. (2021). L-band InSAR depth retrieval during the NASA SnowEx 2020 Campaign: Grand Mesa, Colorado. *Proceedings of the International Geoscience and Remote Sensing Symposium (IGARSS)*
- Lewis, G., Osterberg, E., Hawley, B., **H.-P. Marshall**, Meehan, T., Graeter, K. A., McCarthy, F., Overly, T., Thundercloud, Z., Ferris, D., Koffman, B., and Dibb, J. (2021). Atmospheric blocking drives recent albedo change across the western Greenland ice sheet percolation zone. *Geophysical Research Letters*, 48(10):e2021GL092814
- Johnson, J., Anderson, J., **H.-P. Marshall**, Havens, S., and Watson, L. (2021). Snow avalanche detection and source constraints made using multiple infrasound sensor arrays. *Journal of Geophysical Research-Earth Surface*, 126(3):e2020JF005741
- Liu, J., Enderlin, E. M., **H.P. Marshall**, and Khalil, A. (2021). Automated detection of marine glacier calving fronts using the 2d wavelet transform modulus maxima (wtmm) segmentation method. *IEEE Transactions On Geoscience and Remote Sensing*, 99(DOI: 10.1109/TGRS.2021.3053235):1–10
- Meehan, T. G., **H.P. Marshall**, Bradford, J. H., Hawley, R. L., Overly, T. B., Lewis, G., Graeter, K., Osterburg, E., and McCarthy, F. (2021). Historical surface mass balance reconstruction 1984-2017 from greentracs multi-offset ground-penetrating radar. *Journal of Glaciology*, 67:219–228
- Hojatimalekshah, A., Uhlmann, Z., Glenn, N. F., Hiemstra, C. A., Tennant, C. J., Graham, J. D., Spaete, L., Gelvin, A., **H.P. Marshall**, McNamara, J., and Enterkine, J. (2021). Tree canopy and snow depth relationships at fine scales with terrestrial laser scanning. *The Cryosphere*, 15(doi.org/10.5194/tc-15-2187-2021):2187–2209
- Kim, R. S., Kumar, S., Vuyovich, C., Houser, P., Lundquist, J., Mudryk, L., Durand, M., Barros, A., Kim, E. J., Forman, B. A., Gutmann, E. D., Wrzesien, M. L., Garnaoud, C., Sandells, M., **H.P. Marshall**, Cristea, N., Pflug, J. M., Johnston, J., Cao, Y., Mocko, D., and Wang, S. (2021). Snow ensemble uncertainty project (seup): Quantification of snow water equivalent uncertainty across north america via ensemble land surface modeling. *The Cryosphere*, 15(doi.org/10.5194/tc-15-771-2021):771–791
- Arenson, L., Colgan, W., and **Marshall, H.P.** (2021, 2nd edition). *Physical, Thermal, and Mechanical Properties of Snow, Ice, and Permafrost*, chapter 2, pages 35–75. Snow and ice-related hazards, risks, and disasters. Elsevier, ed. Haeberli and Whiteman, Radarweg 29, PO Box 211, 1000 AE Amsterdam, Netherlands
- Webb, R. W., Marziliano, A., McGrath, D., Bonnell, R., Meehan, T. G., Vuyovich, C., and **H.-P. Marshall** (2021). In situ determination of dry and wet snow permittivity: Improving equations for low frequency radar applications. *Remote Sensing*, 13(22):DOI:10.3390/rs13224617

- Bonnell, R., McGrath, D., Williams, K., Webb, R., Fassnacht, S. R., and **H.-P. Marshall** (2021). Spatiotemporal variations in liquid water content in a seasonal snowpack: Implications for radar remote sensing. *Remote Sensing*, 13(21):DOI:10.3390/rs13214223
- 2020 Hedrick, A. R., Marks, D., **H.P. Marshall**, McNamara, J., Havens, S., Trujillo, E., Sandusky, M., Robertson, M., Johnson, M., Bormann, K. J., and Painter, T. H. (2020). From drought to flood: A water balance analysis of the tuolumne river basin during extreme conditions (2015 – 2017). *Hydrol. Process.*, 34(11):2560–2574
- Webb, R., Raleigh, M., McGrath, D., Molotch, N., Elder, K., Hiemstra, C., Brucker, L., and **H.P. Marshall** (2020). Within-stand boundary effects of snow water equivalent distribution in forested areas. *Water Resources Research*, 56:e2019WR024905
- Aguayo, M. A., Flores, A. N., McNamara, J. P., **H.P. Marshall**, and Mead, J. (2020). Examining cross-scale influences of forcing resolutions in a hillslope-resolving, integrated hydrologic model. *Hydrology and Earth System Sciences Discussions*, (doi.org/10.5194/hess-2020-451)
- 2019 McGrath, D., Webb, R., Shean, D., Bonnell, R., **H.P. Marshall**, Painter, T., Molotch, N., Elder, K., Hiemstra, C., and Brucker, L. (2019). Spatially extensive ground-penetrating radar snow depth observations during nasa’s 2017 snowex campaign: Comparison to in situ, airborne, and satellite observations. *Water Resources Research*, 55(11):10026–10036
- Lievens, H., Demuzere, M., **H.P. Marshall**, Reichle, R. H., Brucker, L., Brangers, I., de Rosnay, P., Dumont, M., Giroto, M., Immerzeel, W. W., Jonas, T., Kim, E. J., Koch, I., Marty, C., Saloranta, T., Schöber, J., and Lannoy, G. J. D. (2019). Snow depth variability in the northern hemisphere mountains observed from space. *Nature Communications*, 10(1):1–12
- Lund, J., Forster, R. R., Rupper, S. B., **H.P. Marshall**, Deeb, E. J., and Hashmi, M. (2019). Mapping snowmelt progression in the upper indus basin with synthetic aperture radar. *Frontiers in Earth Science*, 7:318
- Lewis, G., Osterberg, E., Hawley, R., Marshall, H. P., Meehan, T., Graeter, K., McCarthy, F., Overly, T., Thundercloud, Z., and Ferris, D. (2019a). Recent precipitation decrease across the western greenland ice sheet percolation zone. *The Cryosphere*, 13:2797–2815
- 2018 Graeter, K. A., Osterberg, E. C., Ferris, D. G., Hawley, R. L., **H.P. Marshall**, Lewis, G., Meehan, T., McCarthy, F., and Birkel, S. D. (2018). Ice core records of West Greenland surface melt and climate forcing. *Geophysical Research Letters*, 45(7):3164–3172
- Hedrick, A., Marks, D., Kormos, P., **H.P. Marshall**, Havens, S., Bormann, K., and Painter, T. H. (2018). Direct insertion of NASA Airborne Snow Observatory-derived snow depth time-series into the iSnobal energy balance snow model. *Water Resources Research*, 54(10):8045–8063
- McGrath, D., Sass, L., O’Neel, S., McNeil, C., Candela, S. G., Baker, E. H., and **H.P. Marshall** (2018). Interannual snow accumulation variability on glaciers derived from repeat, spatially extensive ground-penetrating radar surveys. *The Cryosphere Discussions*, 12(doi.org/10.5194/tc-2018-126):3617–3633

- Kim, Y., Reck, T. J., Alonso-delPino, M., Painter, T. H., **H.P. Marshall**, Bair, E. H., Dozier, J., Chattopadhyay, G., Liou, K.-N., Chang, M.-C. F., and Tang, A. (2018). A Ku-band CMOS FMCW radar transceiver for snowpack remote sensing. *IEEE Transactions On Geoscience and Remote Sensing*, 66(5):2480–2494
- Hagenmuller, P., van Herwijnen, A., Pielmeier, C., and **H.P. Marshall** (2018). Evaluation of the snow penetrometer Avatech SP2. *Cold Regions Science and Technology*, 149:83–94
- McNamara, J. P., Benner, S. G., Poulos, M. J., Pierce, J. L., Chandler, D. G., Kormos, P. R., **H.P. Marshall**, Flores, A. N., Seyfried, M., Glenn, N. F., and Aishlin, P. (2018). Form and function relationships revealed by long-term research in a semiarid mountain catchment. *WIREs Water*, e1267(10.1002/wat2.1267)
- 2017 Mikesell, T., van Wijk, K., Otheim, L., **H.P. Marshall**, and Kurbatov, A. (2017). Laser ultrasound observations of mechanical property variations in ice cores. *Geosciences*, 7(3):47
- Lewis, G., Osterberg, E., Hawley, R., Whitmore, B., **H.P. Marshall**, and Box, J. (2017a). Regional Greenland accumulation variability from Operation IceBridge airborne accumulation radar. *The Cryosphere*, 11(2):773
- Pegau, W. S., Garron, J., Zabilansky, L., Bassett, C., Bello, J., Bradford, J., Carns, R., Courville, Z., Eicken, H., Elder, B., Eriksen, P., Lavery, A., Light, B., Maksym, T., **H.P. Marshall**, Oggier, M., Perovich, D., Pacwiardowski, P., Singh, H., Tang, D., Wiggins, C., and Wilkinson, J. (2017). Detection of oil in and under ice. *International Oil Spill Conference Proceedings*, 2017(1):1857–1876
- 2016 Evans, S., Flores, A., Heilig, A., Kohn, M., **Marshall, H.P.**, and McNamara, J. (2016). Isotopic evidence for lateral flow and diffusive transport, but not sublimation, in a sloped seasonal snowpack, Idaho, U.S.A. *Geophysical Research Letters*, 43(7):3298–3306
- Rutter, N., **Marshall, H.P.**, Tape, K., Essery, R., and King, J. (2016). Impact of spatial averaging on radar reflectivity at internal snowpack layer boundaries. *Journal of Glaciology*, 62(236):1065–1074
- Bradford, J. H., Babcock, E. L., **Marshall, H.P.**, and Dickins, D. F. (2016). Targeted reflection-waveform inversion of experimental ground-penetrating radar data for quantification of oil spills under sea ice. *Geophysics*, 81(1):WA59–WA70
- 2015 Heilig, A., Mitterer, C., Schmid, L., Wever, N., Schweizer, J., **Marshall, H.P.**, and Eisen, O. (2015). Seasonal and diurnal cycles of liquid water in snow-measurements and modeling. *Journal of Geophysical Research-Earth Surface*, 120(10):2139–2154
- Kormos, P. R., McNamara, J. P., Seyfried, M. S., **Marshall, H.P.**, Marks, D., and Flores, A. N. (2015). Bedrock infiltration estimates from a catchment water storage-based modeling approach in the rain snow transition zone. *Journal of Hydrology*, 525:231–248
- Hedrick, A., **Marshall, H.P.**, Winstral, A., Elder, K., Yueh, S., and Cline, D. (2015). Independent evaluation of the SNODAS snow depth product using regional-scale LiDAR-derived measurements. *Cryosphere*, 9(1):13–23

Research Funding Summary

Agency	Short Title	Role	Dates	Amount
CRREL	Advancement of snow monitoring	PI	9/2020-8/2022	\$1,000,000
NASA IIP	Meta-surface Radar for Snow Remote Sensing	PI	2/2020-1/2023	\$2,168,639
U.S. DOTs	Enhanced Infrasound Avalanche Detection	PI	6/2017-8/2018	\$99,509
NSF	MRI: Long-range UAV for Sea-Ice Monitoring	Co-PI	9/2018-8/2022	\$1,615,385
ID Dept Commerce	A General Purpose Goniometer	Co-PI	6/2018-7/2020	\$368,772
NASA	Earth Systems Science Fellowship (Gongora)	PI	09/2018-8/2019	\$90,000
ITD	Continuous snow temperature monitoring	PI	9/2018-8/2020	\$45,000
U.S. DOTs	Infrasound Avalanche Monitoring	Co-PI	1/2019 - 12/2022	\$99,509
NASA	NASA SnowEx Ground-based Radar	Co-PI	6/2018-6/2021	\$444,742
NASA	NASA SnowEx Y1 Data Assimilation	Co-PI	6/2018-6/2021	\$421,212
NASA	Spatiotemporal Patterns in GBRs	PI	6/2017-6/2022	\$460,719
NASA	L-Band InSAR for snow depth and SWE	Co-PI	6/2017-6/2022	\$450,000
NASA JPL	ASO and UAVSAR data fusion	PI	6/2017-7/2018	\$21,103
U.S. DOTs	Infrasound Avalanche Detection	PI	6/2017-8/2018	\$99,339
NASA	Broadband Array IIP - Phase 2	PI	02/2015-05/2017	\$330,000
NSF	GreenTRACS	PI	01/2015-12/2017	\$355,338
NASA	Multi-frequency Data Assimilation	Co-PI	07/2015-06/2018	\$295,577
NASA	Monitoring Earth's Hydrosphere	Co-PI	06/2014-05/2017	\$749,870
U.S. Army	SWE estimates using radar phase	Co-PI	10/2013- 9/2016	\$490,800
OGP	Remote Sensing Tests with Oil in Ice	Co-PI	01/2014-12/2014	\$128,739
NASA	NASA Snow School	Co-PI	01/2014-12/2014	\$65,385
NSF	MRI: Laser-ultrasonic Ice Core Tomography	Co-PI	09/2012-08/2013	\$336,122
NSF	MRI:GPU Computing/Visualization Cluster	Co-PI	09/2012-08/2013	\$555,384
NSF	Measuring Sublimation with Isotopes	Co-PI	04/2012-04/2013	\$19,912
ITD	Avalanche Detection	PI	09/2012-08/2015	\$100,000
NASA	WISM IIP - Phase 1	PI	11/2011-12/2014	\$94,947
NASA	Remote Sensing of the Cryosphere	PI	01/2011-12/2013	\$1,124,552
NASA	Improvement of Remotely Sensed SWE	PI	09/2010-08/2013	\$329,953
NSF	Lateral Flow of Water in Snow	PI	04/2010-03/2013	\$228,963
NASA	Earth Systems Science Fellowship (Havens)	PI	09/2010-12/2013	\$90,000
D.F. Dickins	Oil Detection Radar	PI	01/2010-12/2012	\$328,747
IDT	Avalanche Modeling	PI	09/2010-08/2012	\$10,000
NASA	CLPX-II Grand Mesa, Colorado	PI	08/2010-08/2012	\$35,468
NASA	Investigation of Snowpack Properties	PI	08/2006-07/2009	\$462,132
NSF	Radar for Measuring Water Pathways	PI	09/2005-08/2010	\$69,785
			TOTAL	\$13,585,603